The data are from a longitudinal clinical trial of contracepting women. In this trial women received an injection of either 100 mg or 150 mg of depot-medroxyprogesterone acetate (DMPA) on the day of randomization and three additional injections at 90-day intervals. There was a final follow-up visit 90 days after the fourth injection, i.e., one year after the first injection.

Throughout the study each woman completed a menstrual diary that recorded any vaginal bleeding pattern disturbances. The diary data were used to determine whether a women experienced amenorrhea, the absence of menstrual bleeding for a specified number of days.

A total of 1151 women completed the menstrual diaries and the diary data were used to generate a binary sequence for each woman according to whether or not she had experienced amenorrhea in the four successive three month intervals.

Reference: Machin D, Farley T, Busca B, Campbell M and d'Arcangues C. (1988). *Assessing changes in vaginal bleeding patterns in contracepting women*. **Contraception**, 38, 165-179.

```
proc print  data = BIOS755.Amenorrhea (obs=10);
run;
```

The SAS System

Obs	ID	TRT	TIME	Y	Ctime	prevy
1	1	0	1	0	1	.
2	1	0	2	.	2	0
3	1	0	3	.	3	.
4	1	0	4	.	4	.
5	2	0	1	0	1	.
6	2	0	2	.	2	0
7	2	0	3	.	3	.
8	2	0	4	.	4	.
9	3	0	1	0	1	.
10	3	0	2	.	2	0

Below, we'll use a weighted GEE to analyze the data:

First we create a version of the data with the lagged outcome and a second time variable.

```
data BIOS755.Amenorrhea;
set BIOS755.Amenorrhea;
Ctime = time;
prevy = lag(y);
run;
```

Below the `missmodel` command is what estimates the weights.

For the WGEE weight model we are using categorical time (profile analysis), for the actual model we're using continuous time. This is why we needed two time variables.

```
ods graphics on;
proc gee data=BIOS755.Amenorrhea desc plots=histogram;
   class ID Ctime;
   missmodel Ctime Prevy trt trt*Prevy / type=obslevel;
   model Y = Time trt Time*Time trt*Time trt*Time*Time / dist=bin;
   repeated subject=ID / within=Ctime corr=cs;
run;
```

This runs the following model:

$$logit\left(p\left(r_{ij}=1\middle|r_{ij-1}=1\right)\right)=\alpha_0+\alpha_1 I\left(t_{ij}=2\right)+\alpha_2 I\left(t_{ij}=3\right)+\alpha_3 TRT_i+\alpha_4 y_{ij-1}+\alpha_5 TRT_i y_{ij-1}$$

which is used to estimate

$$w_{ij}=\frac{1}{p\left(r_{ij}=1\middle|r_{ij-1}=1\right)}$$

Observation-level weights are used when each measurement has its own weight. This is common when weights vary over time, such as with **inverse probability weights for missing data or dropout**, where the probability of being observed can differ at each time point.

Subject-level weights `(type=sublevel)` are used when the weight applies to the individual as a whole and is constant across all of their observations. This is common with **inverse probability weights for missing baseline covariates**.

Model Information			
Data Set	BIOS755.AMENORRHEA		y
Distribution	Binomial		
Link Function	Logit		
Dependent Variable	Y		

Number of Observations Read	4604
Number of Observations Used	3616
Number of Events	1231

Number of Trials	3616
Number of Missing Values	988

Class Level Information		
Class	**Levels**	**Values**
ID	1151	1 2 3 4 5 6 7 8 9 10 …. 77 78 79 80 81 82 83 84 85 86 87 ...
Ctime	4	1 2 3 4

Response Profile		
Ordered Value	**Y**	**Total Frequency**
1	1	1231
2	0	2385

PROC GEE is modeling the probability that Y='1'.

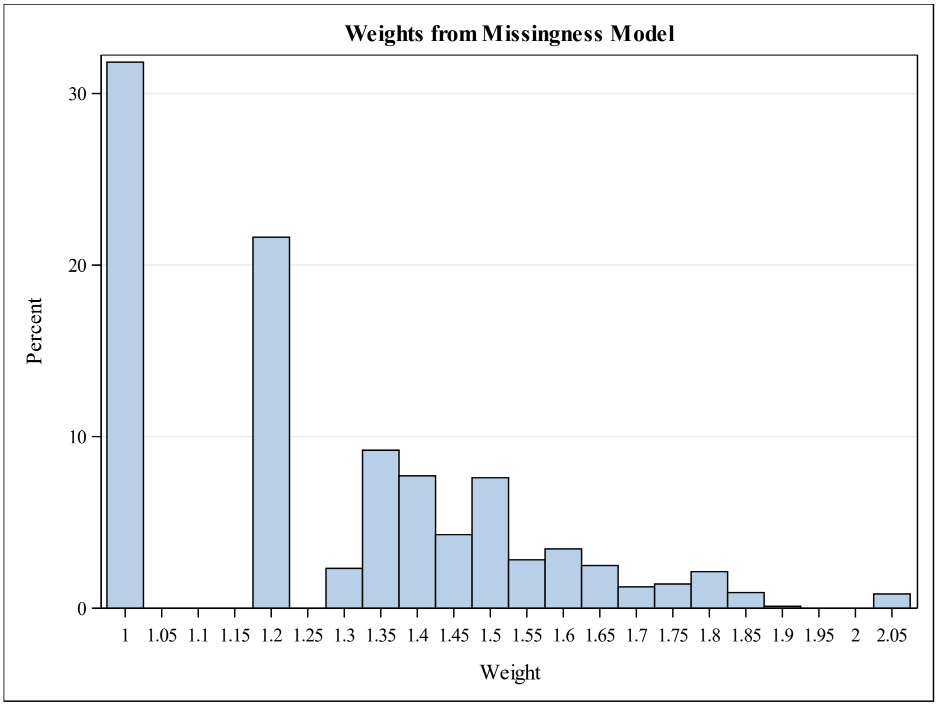
GEE Model Information	
Correlation Structure	Exchangeable
Within-Subject Effect	Ctime (4 levels)
Subject Effect	ID (1151 levels)
Number of Clusters	1151
Clusters With Missing Values	437
Correlation Matrix Dimension	4
Maximum Cluster Size	4
Minimum Cluster Size	1

Missing Data Patterns						
	Occasion					
Group	**1**	**2**	**3**	**4**	**Freq**	**Percent**
1	X	X	X	X	714	62.03
2	X	X	X	.	84	7.30
3	X	X	.	.	155	13.47
4	X	.	.	.	198	17.20

Exchangeable Working Correlation	
Correlation	0.3949

Parameter Estimates for Response Model with Empirical Standard Error Estimates						
Parameter	**Estimate**	**Standard Error**	**95% Confidence Limits**		**Z**	**Pr > |Z|**
Intercept	-2.0381	0.2484	-2.5251	-1.5512	-8.20	<.0001
TIME	0.5453	0.2117	0.1303	0.9603	2.58	0.0100
TRT	-0.4296	0.3540	-1.1234	0.2643	-1.21	0.2250
TIME*TIME	-0.0037	0.0405	-0.0831	0.0757	-0.09	0.9275
TRT*TIME	0.6621	0.3021	0.0700	1.2542	2.19	0.0284
TRT*TIME*TIME	-0.1264	0.0577	-0.2395	-0.0134	-2.19	0.0284

Parameter Estimates for Missingness Model							
Parameter		**Estimate**	**Standard Error**	**95% Confidence Limits**		**Z**	**Pr > |Z|**
Intercept		2.3967	0.1438	2.1149	2.6785	16.67	<.0001
Ctime	1	0.0000	0.0000	0.0000	0.0000	.	.
Ctime	2	-0.7286	0.1439	-1.0106	-0.4466	-5.06	<.0001
Ctime	3	-0.5919	0.1469	-0.8798	-0.3040	-4.03	<.0001
Ctime	4	0.0000	0.0000	0.0000	0.0000	.	.
prevy		-0.4514	0.1619	-0.7687	-0.1341	-2.79	0.0053
TRT		0.0680	0.1313	-0.1893	0.3253	0.52	0.6046
prevy*TRT		-0.2381	0.2196	-0.6685	0.1923	-1.08	0.2782



Comparing with the complete data analysis

```
proc gee data=BIOS755.Amenorrhea desc plots=histogram;
   class ID Ctime;
   model Y = Time trt Time*Time trt*Time trt*Time*Time / dist=bin;
   repeated subject=ID / within=Ctime corr=cs;
run;
```

Model Information			
Data Set	BIOS755.AMENORRHEA		y
Distribution	Binomial		
Link Function	Logit		
Dependent Variable	Y		

Number of Observations Read	4604
Number of Observations Used	3616
Number of Events	1231
Number of Trials	3616
Number of Missing Values	988

Class Level Information		
Class	**Levels**	**Values**
ID	1151	1 2 3 4 ..... 75 76 77 78 79 80 81 82 83 84 85 86 87 ...
Ctime	4	1 2 3 4

Response Profile		
Ordered Value	**Y**	**Total Frequency**
1	1	1231
2	0	2385

PROC GEE is modeling the probability that Y='1'.

GEE Model Information	
Correlation Structure	Exchangeable
Within-Subject Effect	Ctime (4 levels)

GEE Model Information	
Subject Effect	ID (1151 levels)
Number of Clusters	1151
Clusters With Missing Values	437
Correlation Matrix Dimension	4
Maximum Cluster Size	4
Minimum Cluster Size	1

Exchangeable Working Correlation	
Correlation	0.3626

GEE Fit Criteria	
QIC	4384.4585
QICu	4382.9860

Parameter Estimates for Response Model with Empirical Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-2.0022	0.2466	-2.4856	-1.5187	-8.12	<.0001
TIME	0.5097	0.2095	0.0990	0.9205	2.43	0.0150
TRT	-0.4516	0.3514	-1.1403	0.2371	-1.29	0.1987
TIME*TIME	0.0014	0.0401	-0.0773	0.0801	0.03	0.9724
TIME*TRT	0.6902	0.2988	0.1045	1.2760	2.31	0.0209
TIME*TIME*TRT	-0.1299	0.0571	-0.2419	-0.0180	-2.27	0.0229

Here's the weighted estimates:

Parameter Estimates for Response Model with Empirical Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-2.0381	0.2484	-2.5251	-1.5512	-8.20	<.0001
TIME	0.5453	0.2117	0.1303	0.9603	2.58	0.0100
TRT	-0.4296	0.3540	-1.1234	0.2643	-1.21	0.2250
TIME*TIME	-0.0037	0.0405	-0.0831	0.0757	-0.09	0.9275
TRT*TIME	0.6621	0.3021	0.0700	1.2542	2.19	0.0284
TRT*TIME*TIME	-0.1264	0.0577	-0.2395	-0.0134	-2.19	0.0284

Even though the lagged outcome is associated with missingness (suggesting MAR), this does not necessarily imply that the unweighted GEE is biased in a meaningful way. If the variables driving missingness (e.g., the lag-1 outcome) are already strongly related to the outcome and partially captured by the mean model structure (through time and treatment), then the bias in the unweighted GEE may be small. In that case, the weighted GEE is correcting for a mechanism that does not substantially distort the estimates, so both approaches yield similar results.

Creating monotone missing data in SAS data step

First, read in the standard data file

```
proc import file= 'C:/Users/mclaina/OneDrive - University of South
Carolina/Teaching/755_Spring_2022/Homework/ahead.csv' out=ahead dbms=csv
replace;
getnames=Yes;
run;
```

Now, create lag variable to use in missingness model:

```
data ahead_lag;
set ahead;
```

```
iadlany_lag = lag(iadlany);
run;

proc gee data=ahead_lag desc plots=histogram;
   class id;
   missmodel some missingness model / type=obslevel;
   model iadlany = some model  / dist=bin;
   repeated subject=id / corr=cs;
run;
```

```

proc gee data=ahead_lag desc plots=histogram;
  class id;
  missmodel some missingness model / type=obslevel;
  model iadlany = some model  / dist=bin;
  repeated subject=id / corr=cs;
run;
```

ERROR: Non-monotone missingness is not allowed.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE GEE used (Total process time):
    real time   0.21 seconds
    cpu time   0.18 seconds

The SAS System

The GEE Procedure

Model Information

Data Set	WORK.AHEAD_LAG
Distribution	Binomial
Link Function	Logit
Dependent Variable	iadlany

Missing Data Patterns

Group	Occasion				Freq	Percent
	1	2	3	4		
1	X	X	X	X	3814	57.34
2	X	X	X	.	872	13.11
3	X	X	.	X	79	1.19
4	X	X	.	.	925	13.91
5	X	.	X	X	90	1.35
6	X	.	X	.	34	0.51
7	X	.	.	X	28	0.42

Missing Data Patterns

Group	Occasion				Freq	Percent
	1	2	3	4		
8	X	.	.	.	809	12.16

```
data ahead_mono;
set ahead_lag;
iadlany_mono = .;
if iadlany_lag ne . then iadlany_mono=iadlany;
if year = 0 then iadlany_mono=iadlany;
run;

proc gee data=ahead_mono desc plots=histogram;
   class id;
   missmodel some missingness model / type=obslevel;
   model iadlany_mono = some model  / dist=bin;
   repeated subject=id / corr=cs;
run;
```

The SAS System

The GEE Procedure

Model Information

Data Set	WORK.AHEAD_MONO
Distribution	Binomial
Link Function	Logit
Dependent Variable	iadlany_mono

Missing Data Patterns

Group	Occasion				Freq	Percent
	1	2	3	4		
1	X	X	X	X	3814	57.34
2	X	X	X	.	872	13.11
3	X	X	.	.	1004	15.10
4	X	.	.	X	90	1.35
5	X	.	.	.	871	13.10

```
data ahead_mono2;
set ahead_lag;
```

```
by id;
  miss = 0;
  do i=1 to 4;
  if first.id then miss = 0;
  if iadlany eq . then miss = 1;
  if lag(miss) eq 1 then miss = 1;
  iadlany_mono = iadlany;
  if miss = 1 then iadlany_mono = .;
  end;
  output;
drop i miss;
run;


proc gee data=ahead_mono2 desc plots=histogram;
   class id;
   missmodel some missingness model / type=obslevel;
   model iadlany_mono = some model  / dist=bin;
   repeated subject=id / corr=cs;
run;
```

The SAS System

The GEE Procedure

Model Information

Data Set	WORK.AHEAD_MONO2
Distribution	Binomial
Link Function	Logit
Dependent Variable	iadlany_mono

Number of Observations Read	26604
Number of Observations Used	20841
Number of Events	5555
Number of Trials	20841
Number of Missing Values	5763

Class Level Information

Class	Levels	Values
id	6651	1 10 100 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 101 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 102 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 103 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 104 1040 1041 1042 1043 1044 ...

Response Profile

Ordered Value	iadlany_mono	Total Frequency
1	1	5555
2	0	15286

PROC GEE is modeling the probability that iadlany_mono='1'.

GEE Model Information

Correlation Structure	Exchangeable
Subject Effect	id (6651 levels)
Number of Clusters	6651
Clusters With Missing Values	2837
Correlation Matrix Dimension	4
Maximum Cluster Size	4
Minimum Cluster Size	1

Missing Data Patterns

Group	Occasion				Freq	Percent
	1	2	3	4		
1	X	X	X	X	3814	57.34
2	X	X	X	.	872	13.11
3	X	X	.	.	1004	15.10
4	X	.	.	.	961	14.45

Multiple Imputation in SAS

To impute longitudinal outcome data we use the monotone statement on the wide version of the dataset.

```
proc transpose data=BIOS755.Amenorrhea out=Amenorrhea_wide prefix=Y;
by id;
id time;
var y;
run;

data BIOS755.Amenorrhea_wide;
merge BIOS755.Amenorrhea Amenorrhea_wide;
by id;
if first.id;
drop y _NAME_ _LABEL_;
run;

proc print  data = BIOS755.Amenorrhea_wide (obs=10);
run;
```

Obs	ID	TRT	TIME	Y1	Y2	Y3	Y4
1	1	0	1	0	.	.	.
2	2	0	1	0	.	.	.
3	3	0	1	0	.	.	.

```
proc sort data = BIOS755.Amenorrhea_wide;
by TRT;
run;
```

With a MONOTONE statement, the variables with missing values are imputed sequentially in the order specified in the VAR statement.

```
proc mi data=BIOS755.Amenorrhea_wide NIMPUTE=10 seed=8675309 out=MI_wide;
* We could use the "by" statement to impute separately for each ID.
* We don't want to do that here, but it's an option when there is lots of
data per person/cluster;
* by id;
class y1 - y4;
var trt y1 - y4;
monotone logistic (y2 = y1 trt);
monotone logistic (y3 = y2 y1 y1*y2 trt);
monotone logistic (y4 = y3 y2 y1 y1*y2 y3*y2 trt);
* We don’t specify link = logit, but that is what is used (by default).
With multinomial data,
    •  link = logit will use the cumulative logistic model for ordinal data,
    •  link = glogit for nominal data;
run;
```

Model Information	
Data Set	BIOS755.AMENORRHEA_WIDE
Method	Monotone
Number of Imputations	10
Seed for random number generator	8675309

Monotone Model Specification	
Method	**Imputed Variables**
Logistic Regression	Y2 Y3 Y4
Discriminant Function	Y1

Missing Data Patterns								
								Group Means
Group	**TRT**	**Y1**	**Y2**	**Y3**	**Y4**	**Freq**	**Percent**	**TRT**
1	X	X	X	X	X	714	62.03	0.494398
2	X	X	X	X	.	84	7.30	0.428571
3	X	X	X	.	.	155	13.47	0.561290
4	X	X	.	.	.	198	17.20	0.500000

Then, I'll do the same things I did before. Comparing the means of the imputed variables we the means of the observed variables.

Obs	TRT	NImpute	Parm	Estimate	StdErr	LCLMean	UCLMean	DF	Min	Max	Theta0	tValue	Probt
1	0	10	mn_Y1	0.18576	0.01622	.	.	.	0.18576	0.18576	0	.	.
2	0	10	mn_Y2	0.26892	0.01986	0.2299	0.30795	505.4	0.26042	0.27951	0	13.54	<.0001
3	0	10	mn_Y3	0.39983	0.02587	0.34815	0.45151	63.42	0.37326	0.42361	0	15.46	<.0001
4	0	10	mn_Y4	0.51962	0.02529	0.46936	0.56988	87.05	0.49653	0.54167	0	20.55	<.0001
5	1	10	mn_Y1	0.20522	0.01686	.	.	.	0.20522	0.20522	0	.	.
6	1	10	mn_Y2	0.34661	0.02164	0.30405	0.38917	360.9	0.33391	0.35652	0	16.01	<.0001
7	1	10	mn_Y3	0.51478	0.02513	0.46488	0.56469	92.81	0.49044	0.5287	0	20.48	<.0001
8	1	10	mn_Y4	0.5673	0.02477	0.51815	0.61646	97.76	0.54087	0.58435	0	22.9	<.0001

The MEANS Procedure

trt=0

Variable	Mean	Std Dev	Lower 95% CL for Mean	Upper 95% CL for Mean
Y1	0.1857639	0.3892541	0.1539083	0.2176194
Y2	0.2620545	0.4402138	0.2224488	0.3016603
Y3	0.3887531	0.4880641	0.3413121	0.4361940
Y4	0.5013850	0.5006920	0.4495614	0.5532087

trt=1

Variable	Mean	Std Dev	Lower 95% CL for Mean	Upper 95% CL for Mean
Y1	0.2052174	0.4042120	0.1721088	0.2383259
Y2	0.3361345	0.4728825	0.2935446	0.3787243
Y3	0.4935733	0.5006026	0.4436707	0.5434759
Y4	0.5354108	0.4994524	0.4831289	0.5876926

Go from wide to long:

```
data MI_wide_long;
set MI_wide;
  time=1;
  Y=Y1;
  output;
  time=2;
  Y=Y2;
  output;
  time=3;
  Y=Y3;
  output;
  time=4;
  Y=Y4;
  output;
  drop Y1-Y4;
run;
run;
```

Now, we'll analyze the data using proc glimmix by _Imputation_:

```
proc glimmix data=MI_wide_long method=QUAD(qpoints=50);
   by _Imputation_;
   class ID;
   model Y = Time trt Time*Time trt*Time trt*Time*Time / dist=bin solution
covb;
   random intercept  / subject=id;
   ods output ParameterEstimates=mixparms CovB=mixcovb;
run;
```

The output from the first 2 imputations:

The SAS System

The GLIMMIX Procedure
Imputation Number=1

Model Information

Data Set	WORK.MI_WIDE_LONG
Response Variable	Y
Response Distribution	Binomial
Link Function	Logit
Variance Function	Default
Variance Matrix Blocked By	ID
Estimation Technique	Maximum Likelihood
Likelihood Approximation	Gauss-Hermite Quadrature
Degrees of Freedom Method	Containment

Class Level Information

Class	Levels	Values
ID	1151	1 2 3 4 5 6 7 8 9 ……. 1151

Number of Observations Read	4604
Number of Observations Used	4604

Dimensions

G-side Cov. Parameters	1
Columns in X	6

Dimensions

Columns in Z per Subject	1
Subjects (Blocks in V)	1151
Max Obs per Subject	4

Optimization Information

Optimization Technique	Dual Quasi-Newton
Parameters in Optimization	7
Lower Boundaries	1
Upper Boundaries	0
Fixed Effects	Not Profiled
Starting From	GLM estimates
Quadrature Points	50

Iteration History

Iteration	Restarts	Evaluations	Objective Function	Change	Max Gradient
0	**0**	4	5076.0941281	.	1097.241
1	**0**	4	5067.7387036	8.35542455	201.1249
2	**0**	2	5049.0448135	18.69389005	726.9587
3	**0**	3	5029.056989	19.98782452	192.037
14	**0**	3	4963.0965641	0.00157926	0.169368
15	**0**	3	4963.096537	0.00002709	0.015629

Convergence criterion (GCONV=1E-8) satisfied.

Fit Statistics

-2 Log Likelihood	4963.10
AIC (smaller is better)	4977.10
AICC (smaller is better)	4977.12
BIC (smaller is better)	5012.44
CAIC (smaller is better)	5019.44

Fit Statistics

HQIC (smaller is better)	4990.44

Fit Statistics for Conditional Distribution

-2 log L(Y | r. effects)	2950.53
Pearson Chi-Square	2486.32
Pearson Chi-Square / DF	0.54

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error
Intercept	**ID**	4.8233	0.4605

Solutions for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	-3.5029	0.3822	1149	-9.16	<.0001
TIME	0.9244	0.3135	3449	2.95	0.0032
TRT	-0.6139	0.5237	3449	-1.17	0.2412
TIME*TIME	-0.00429	0.06015	3449	-0.07	0.9432
TIME*TRT	1.0102	0.4383	3449	2.30	0.0213
TIME*TIME*TRT	-0.1831	0.08420	3449	-2.17	0.0298

Type III Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
TIME	1	3449	8.70	0.0032
TRT	1	3449	1.37	0.2412
TIME*TIME	1	3449	0.01	0.9432
TIME*TRT	1	3449	5.31	0.0213
TIME*TIME*TRT	1	3449	4.73	0.0298

Covariance Matrix for Fixed Effects

Effect	Row	Col1	Col2	Col3	Col4	Col5	Col6
Intercept	**1**	0.1461	-0.1090	-0.1392	0.01924	0.1060	-0.01897
TIME	**2**	-0.1090	0.09826	0.1075	-0.01850	-0.09762	0.01844
TRT	**3**	-0.1392	0.1075	0.2743	-0.01929	-0.2112	0.03785
TIME*TIME	**4**	0.01924	-0.01850	-0.01929	0.003618	0.01852	-0.00362
TIME*TRT	**5**	0.1060	-0.09762	-0.2112	0.01852	0.1921	-0.03627
TIME*TIME*TRT	**6**	-0.01897	0.01844	0.03785	-0.00362	-0.03627	0.007090

The GLIMMIX Procedure
Imputation Number=2

Model Information

Data Set	WORK.MI_WIDE_LONG
Response Variable	Y
Response Distribution	Binomial
Link Function	Logit
Variance Function	Default
Variance Matrix Blocked By	ID
Estimation Technique	Maximum Likelihood
Likelihood Approximation	Gauss-Hermite Quadrature
Degrees of Freedom Method	Containment

Class Level Information

Class	Levels	Values
ID	1151	1 2 3 4 5 6…. 1151

Number of Observations Read	4604
Number of Observations Used	4604

Dimensions

G-side Cov. Parameters	1
Columns in X	6
Columns in Z per Subject	1
Subjects (Blocks in V)	1151
Max Obs per Subject	4

Optimization Information

Optimization Technique	Dual Quasi-Newton
Parameters in Optimization	7
Lower Boundaries	1

Optimization Information

Upper Boundaries	0
Fixed Effects	Not Profiled
Starting From	GLM estimates
Quadrature Points	50

Iteration History

Iteration	Restarts	Evaluations	Objective Function	Change	Max Gradient
0	**0**	4	5002.670221	.	989.6732
1	**0**	4	4995.2095204	7.46070063	208.1282
10	**0**	3	4873.2681437	0.00026587	0.172507
11	**0**	4	4873.2677845	0.00035921	2.552924

Convergence criterion (GCONV=1E-8) satisfied.

Fit Statistics

-2 Log Likelihood	4873.27
AIC (smaller is better)	4887.27
AICC (smaller is better)	4887.29
BIC (smaller is better)	4922.61
CAIC (smaller is better)	4929.61
HQIC (smaller is better)	4900.61

Fit Statistics for Conditional Distribution

-2 log L(Y | r. effects)	2815.05
Pearson Chi-Square	2376.58
Pearson Chi-Square / DF	0.52

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error
Intercept	**ID**	5.4393	0.5186

Solutions for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	-3.3388	0.3893	1149	-8.58	<.0001
TIME	0.6620	0.3198	3449	2.07	0.0385
TRT	-1.0295	0.5360	3449	-1.92	0.0549
TIME*TIME	0.04142	0.06155	3449	0.67	0.5010
TIME*TRT	1.4343	0.4483	3449	3.20	0.0014
TIME*TIME*TRT	-0.2567	0.08612	3449	-2.98	0.0029

Type III Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
TIME	1	3449	4.28	0.0385
TRT	1	3449	3.69	0.0549
TIME*TIME	1	3449	0.45	0.5010
TIME*TRT	1	3449	10.24	0.0014
TIME*TIME*TRT	1	3449	8.88	0.0029

Covariance Matrix for Fixed Effects

Effect	Row	Col1	Col2	Col3	Col4	Col5	Col6
Intercept	**1**	0.1515	-0.1124	-0.1441	0.01989	0.1091	-0.01957
TIME	**2**	-0.1124	0.1023	0.1113	-0.01931	-0.1018	0.01926
TRT	**3**	-0.1441	0.1113	0.2873	-0.02004	-0.2203	0.03947
TIME*TIME	**4**	0.01989	-0.01931	-0.02004	0.003788	0.01938	-0.00379
TIME*TRT	**5**	0.1091	-0.1018	-0.2203	0.01938	0.2010	-0.03795
TIME*TIME*TRT	**6**	-0.01957	0.01926	0.03947	-0.00379	-0.03795	0.007417

Now, we'll pool the imputed analyses into one set of parameter estimates with proc mialyze:

```
   proc mianalyze parms=mixparms covb(effectvar=rowcol)=mixcovb ;
   *The covb is only needed for multi-variate results (i.e., type III tests
or contrast statements);
      modeleffects Intercept Time trt Time*Time trt*Time trt*Time*Time;
     title 'MIANALYZE Results';
   run;
```

MIANALYZE Results

The MIANALYZE Procedure

Model Information

PARMS Data Set	WORK.MIXPARMS
COVB Data Set	WORK.MIXCOVB
Number of Imputations	10

Variance Information (10 Imputations)

Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency
	Between	Within	Total				
Intercept	0.023032	0.143971	0.169306	401.93	0.175972	0.153840	0.984849
Time	0.025113	0.097413	0.125037	184.4	0.283573	0.229240	0.977590
trt	0.042740	0.272122	0.319136	414.7	0.172769	0.151400	0.985086
Time*Time	0.000937	0.003597	0.004628	181.35	0.286626	0.231205	0.977402
Time*trt	0.058482	0.191145	0.255475	141.94	0.336551	0.262130	0.974457
Time*Time*trt	0.002635	0.007060	0.009959	106.25	0.410524	0.304022	0.970495

Parameter Estimates (10 Imputations)

Parameter	Estimate	Std Error	95% Confidence Limits		DF	Minimum	Maximum	Theta0	t for H0: Parameter =Theta0	Pr > |t|
Intercept	-3.374046	0.411468	-4.18294	-2.56515	401.93	-3.680824	-3.104243	0	-8.20	<.0001
Time	0.821449	0.353606	0.12382	1.51908	184.4	0.627280	1.182889	0	2.32	0.0213
trt	-0.769218	0.564921	-1.87968	0.34125	414.7	-1.029541	-0.426314	0	-1.36	0.1741
Time*Time	0.014475	0.068028	-0.11975	0.14870	181.35	-0.058471	0.043144	0	0.21	0.8317
Time*trt	1.170986	0.505445	0.17181	2.17016	141.94	0.757193	1.460007	0	2.32	0.0219
Time*Time*trt	-0.222907	0.099794	-0.42075	-0.02506	106.25	-0.301403	-0.134900	0	-2.23	0.0276

Comparing with the complete data analysis

```
proc glimmix data=BIOS755.Amenorrhea method=QUAD(qpoints=50);
   class ID;
   model Y = Time trt Time*Time trt*Time trt*Time*Time / dist=bin solution;
   random intercept  / subject=id;
run;
```

Model Information	
Data Set	BIOS755.AMENORRHEA
Response Variable	Y
Response Distribution	Binomial
Link Function	Logit
Variance Function	Default
Variance Matrix Blocked By	ID
Estimation Technique	Maximum Likelihood
Likelihood Approximation	Gauss-Hermite Quadrature
Degrees of Freedom Method	Containment

Class Level Information		
Class	**Levels**	**Values**
ID	1151	1 …. 1150 1151

Number of Observations Read	4604
Number of Observations Used	3616

Dimensions	
G-side Cov. Parameters	1
Columns in X	6
Columns in Z per Subject	1
Subjects (Blocks in V)	1151
Max Obs per Subject	4

Optimization Information	
Optimization Technique	Dual Quasi-Newton
Parameters in Optimization	7
Lower Boundaries	1
Upper Boundaries	0
Fixed Effects	Not Profiled
Starting From	GLM estimates
Quadrature Points	50

Iteration History					
Iteration	**Restarts**	**Evaluations**	**Objective Function**	**Change**	**Max Gradient**
0	**0**	4	3961.1107127	.	754.8834
1	**0**	4	3954.5188083	6.59190441	170.3103
2	**0**	2	3935.2222761	19.29653216	501.6977
3	**0**	3	3916.3366324	18.88564370	125.6327
4	**0**	2	3895.3759089	20.96072351	37.89303
5	**0**	4	3875.3319661	20.04394273	102.5231
6	**0**	3	3867.4836672	7.84829892	86.57043
7	**0**	3	3867.1121884	0.37147880	3.268637
8	**0**	3	3866.9314898	0.18069861	35.01058
9	**0**	3	3866.8928583	0.03863151	3.791026
10	**0**	3	3866.8913504	0.00150794	0.307824

Iteration History					
Iteration	**Restarts**	**Evaluations**	**Objective Function**	**Change**	**Max Gradient**
11	**0**	2	3866.8899544	0.00139600	1.891312
12	**0**	2	3866.8877416	0.00221279	0.181709
13	**0**	3	3866.8874462	0.00029532	0.43449
14	**0**	3	3866.8874348	0.00001144	0.012497

Convergence criterion (GCONV=1E-8) satisfied.

Fit Statistics	
-2 Log Likelihood	3866.89
AIC (smaller is better)	3880.89
AICC (smaller is better)	3880.92
BIC (smaller is better)	3916.23
CAIC (smaller is better)	3923.23
HQIC (smaller is better)	3894.23

Fit Statistics for Conditional Distribution	
-2 log L(Y | r. effects)	2169.87
Pearson Chi-Square	1741.85
Pearson Chi-Square / DF	0.48

Covariance Parameter Estimates			
Cov Parm	**Subject**	**Estimate**	**Standard Error**
Intercept	ID	5.0798	0.5855

Solutions for Fixed Effects					
Effect	**Estimate**	**Standard Error**	**DF**	**t Value**	**Pr > |t|**
Intercept	-3.3943	0.4146	1149	-8.19	<.0001
TIME	0.7978	0.3551	2461	2.25	0.0248
TRT	-0.8049	0.5636	2461	-1.43	0.1534
TIME*TIME	0.01966	0.06968	2461	0.28	0.7779
TIME*TRT	1.2194	0.4978	2461	2.45	0.0144
TIME*TIME*TRT	-0.2296	0.09770	2461	-2.35	0.0188

Type III Tests of Fixed Effects				
Effect	**Num DF**	**Den DF**	**F Value**	**Pr > F**
TIME	1	2461	5.05	0.0248
TRT	1	2461	2.04	0.1534
TIME*TIME	1	2461	0.08	0.7779
TIME*TRT	1	2461	6.00	0.0144
TIME*TIME*TRT	1	2461	5.52	0.0188